

Exercise 1: Inventory Management System

CODE:

```
import java.util.HashMap;

class Product {
    int productId;
    String productName;
    int quantity;
    double price;

    Product(int productId, String productName, int quantity, double price) {
        this.productId = productId;
        this.productName = productName;
        this.quantity = quantity;
        this.price = price;
    }

    void display() {
        System.out.println(productId + " " + productName + " " + quantity + " " + price);
    }
}

public class Main {
    static HashMap<Integer, Product> inventory = new HashMap<>();

    static void addProduct(Product p) {
        inventory.put(p.productId, p);
    }

    static void updateProduct(int id, int quantity, double price) {
        Product p = inventory.get(id);
        if (p != null) {
            p.quantity = quantity;
            p.price = price;
        }
    }
}
```

```

    }

    static void deleteProduct(int id) {
        inventory.remove(id);
    }

    static void displayInventory() {
        System.out.println("ID\tName\t\tQuantity\tPrice");
        for (Product p : inventory.values()) {
            System.out.println(p.productId + "\t" + p.productName + "\t\t" + p.quantity + "\t\t" +
p.price);
        }
    }

    public static void main(String[] args) {
        addProduct(new Product(101, "Laptop", 15, 65000));
        addProduct(new Product(102, "Mouse", 50, 500));
        addProduct(new Product(103, "Keyboard", 30, 1200));
        System.out.println("Initial Inventory");
        displayInventory();
        updateProduct(102, 60, 550);
        deleteProduct(103);
        System.out.println("\nUpdated Inventory");
        displayInventory();
    }
}

```

OUTPUT:

```

Initial Inventory
ID Name      Quantity  Price
101 Laptop    15       65000.0
102 Mouse     50       500.0
103 Keyboard  30       1200.0

Updated Inventory
ID Name      Quantity  Price
101 Laptop    15       65000.0
102 Mouse     60       550.0

```